Validation

Validation never depends on manually traced Chhota Shigri boundaries. It uses four independent
lines of evidence (A–D below), plus probability calibration and co-registration checks.

A. Held-out benchmark tiles (Maslov et al. 2025, official split)

- **Data:** HMA test tiles from the NIRD archive (doi:10.11582/2024.00168). These were never used
  for training or checkpoint selection. No tile lies within 0.05° of Chhota Shigri.
- **Metrics** (`src/validation/benchmark_eval.py`): pooled IoU (Σ TP over Σ TP+FP+FN, as in the
  paper), precision, recall, F1, mean tile IoU, and boundary F1 with a 2 px tolerance. Padding and unlabelled pixels are excluded.
- **Uncertainty:** paired bootstrap over tiles (2000 resamples), 95 % CI of the pooled-IoU difference.

A1. Transfer-learning hierarchy (57 HMA test tiles)

Model	Pooled IoU	Precision	Recall	F1	Boundary F1
M0 global	0.678	0.898	0.735	0.808	0.266
M1 official (global → HMA, authors)	0.748	0.905	0.811	0.856	0.321
M1-local (global → HMA, this work, ≈1.7 % of official steps)	0.758	0.886	0.841	0.863	0.335
M2 (M1-official → Chhota Shigri self-training + HMA replay)	0.779	0.882	0.870	0.876	0.381

Paired difference	ΔIoU	95 % CI
M1-official − M0	+0.070	[+0.040, +0.107]
M1-local − M0	+0.081	[+0.047, +0.125]
M1-local − M1-official	+0.011	[−0.001, +0.024] (not significant)
M2 − M1-official	+0.031	[+0.019, +0.046]
M2 − M0	+0.101	[+0.064, +0.148]

- M1-official's IoU of 0.748 agrees with the paper's ">0.75 for debris-rich HMA". This is an independent check of our inference implementation (normalisation, padding, blending).
- M2's HMA gain cannot be attributed to Chhota Shigri adaptation alone, because M2 also saw
  40 HMA replay tiles. It shows that self-training caused **no forgetting**.

A2. Input ablation (22 HMA test tiles that carry Sentinel-1 VV; equal budgets, same 66 train tiles)

Inputs	Pooled IoU	Precision	Recall	Boundary F1
optical	0.676	0.904	0.729	0.327
optical + DEM	**0.824**	0.918	0.889	0.383
optical + DEM + SAR (VV asc/desc)	0.808	0.915	0.874	0.350

Paired difference	ΔIoU	95 % CI
+ DEM (opt+DEM − opt)	+0.149	[+0.078, +0.241]
+ SAR (opt+DEM+SAR − opt+DEM)	−0.016	[−0.029, −0.008]

The DEM is the most important input after the optical bands. SAR showed no benefit under a reduced, transfer-learning budget: the new SAR branch was randomly initialised and trained for 1000 steps. This does not contradict the paper, where SAR helped with full training.

B. Existing outlines (inventory agreement, not accuracy)

`src/validation/site_validation.py` compares each automated mask with the 2002 RGI7 outline. It
reports IoU over the domain and over the upper glacier (> 5000 m), where 2002–2025 margin change
should be small. The 2002 outline is **not** ground truth for 2017–2025, so this measures agreement, not accuracy. All GLIMS outlines are older than the Sentinel-2 era (2000–2009) and use different divides (11.6–16.8 km²).

C. Published areas

Source	Area (km²)	Epoch / definition
Wagnon et al. 2007, J. Glaciol. 53(183)	15.7	c. 2003, including tributaries
Mandal et al. 2020, J. Glaciol. 66(259)	15.5	glacier-wide, 2002–2019 study period

Source	Area (km²)	Epoch / definition
RGI 7.0 / GAMDAM	16.77	2002-08-02
ICIMOD (GLIMS 154108)	11.65	2009-10-25, different divides

Published retreat for comparison: about 800 m between 1988 and 2003, 53 m a^{-1} (Kulkarni et al. 2007, cited in Wagnon et al. 2007); 800–850 m between 1972 and 2009 (cited in Gantayat et al. 2017). Debris cover rose from 3.4 % (2005) to about 11 % (2014) (Garg et al. 2017, cited in Mandal et al. 2020).

D. Temporal consistency

Reported per model:

- consecutive-epoch IoU,
- area of "unstable" pixels (lost, then re-appeared),
- number of epoch-to-epoch area gains above 2σ,
- number of terminus advances above 2σ.

A physically consistent series on a retreating glacier should show few of each.

Probability calibration

The raw GlaViTU softmax is compressed (all pixels within about 0.24–0.89; label smoothing plus
focal loss). Isotonic calibrators are fitted on HMA val tiles (`src/validation/calibration.py`). For M1-official the held-out ECE goes from 0.239 (raw) to 0.011 (calibrated). Calibrated probabilities are used for (i) the pseudo-label confidence thresholds and (ii) the probability-threshold sensitivity of the area. Binary masks keep the official argmax/0.5 rule.

Co-registration

Phase correlation on stable terrain against 2023-09-11
(`outputs/statistics/coregistration.csv`).
Two scenes exceeded 0.25 px and were Fourier-shifted: 2017-08-23 (0.67 px → residual 0.11 px)
and 2019-09-07 (0.36 px → 0.14 px). The maximum residual per year is ≤ 1.6 m (≤ 0.16 px), within the < 0.5 px target. One scene (2023-08-27) had too little clear stable terrain to measure.

Avoiding circular validation

- No Chhota Shigri pixel label from the model is used to evaluate the same model.

- M2 was trained only on even-year scenes and selected on HMA val. The analysis epochs are odd
  years.
- Benchmark test tiles are never used for selection.
- The analysis-window revision (15 Aug–15 Sep) was made after diagnostics and is reported with
  the full-window sensitivity run.